

EduVerify — MAHE Production Deployment Plan

Document: Step-by-step production go-live after UAT sign-off
Application: EduVerify Document Pre-Verification Portal
Production URL: <https://maheblreduverify.manipal.edu>
Subscription: MAHEBLR-Admission
SPOC: abhijit.das@manipal.edu
Scale: 5,000–10,000 students per admission cycle
Prepared: July 2026

Prerequisite: Complete UAT on MAHE-CI-D0C-UAT-APP01 with signed checklist (bulk upload, student upload, verifier workflow, storage blobs, health check, GitHub deploy green). Do **not** deploy production until UAT is signed off by Verification Cell + IT.

1. Production architecture



2. Production resources (MAHE)

Resource	Name	Purpose
App Service	MAHE-CI-D0C-PRD-APP01	Web + API + frontend
Resource group	MAHE-CI-D0C-PRD-RG01	(confirm exact name in portal)
PostgreSQL	mahe-ci-doc-prd-psqlsvr01	Student & verification metadata
Storage	maheciodocprdstgacct01	Document files (PDF/images)
Key Vault	MAHE-CI-D0C-PRD-KV01	Production secrets
Public URL	maheblreduverify.manipal.edu	After App Gateway + DNS

Managed Identity	97fb7c66-1571-43bb-96dd-814578216a4f	Key Vault / storage access
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3. UAT sign-off checklist (required before Prod)

#	Test	Pass	Sign-off
1	UAT /api/health returns "db":"up"	☐	
2	Admin login ADM-001 works	☐	
3	Bulk upload 50+ students	☐	
4	10 students: login, upload, self-verify	☐	
5	Verifier: preview, verify, ZIP download	☐	
6	Blobs appear in UAT storage container	☐	
7	Verify schedule generate + allocate (if used)	☐	
8	GitHub Deploy MAHE UAT workflow green	☐	
9	Verification Cell approves go-live	☐	
10	MAHE IT approves Prod window	☐	

4. Phase A — Prepare production PostgreSQL

1 Activate PIM Contributor on subscription MAHEBLR-Admission (Azure Portal → My roles → Activate).

2 Open PostgreSQL server: Portal → search `mahe-ci-doc-prd-psqlsvr01` → Overview.
Confirm Status = **Ready**.

3 Create application database: Left menu → **Databases** → **+ Add** → Name: `eduverify` → Save.
Do not use the default `postgres` database for the application.

4 Networking: PostgreSQL → **Networking** → Enable **Allow public access from any Azure service within Azure to this server** → Save.
(After go-live, infra may restrict to private endpoints per target architecture.)

- 5 **Build DATABASE_URL** for production App Service. Password must be URL-encoded (@ → %40):

```
postgresql://psqladmin:YOUR_URL_ENCODED_PASSWORD@mahe-ci-doc-prd-psqlsvr01.postgres.databases.azure.com:5432/eduverify?sslmode=require
```

Store this in Key Vault secret DATABASE-URL — not in email or git.

5. Phase B — Prepare production Blob Storage

- 1 Portal → mahecidocprdstgacct01 → **Containers**.

- 2 Create container eduverify-documents if missing. **Public access level: Private**.

- 3 Store production storage connection string in Key Vault as AZURE-STORAGE-CONNECTION-STRING (or use Managed Identity + Blob Data Contributor role — preferred long-term).

Never use UAT storage (mahecidocuaststgacct01) on production App Service. UAT and Prod storage must remain separate.

6. Phase C — Key Vault secrets (production)

Work with MAHE IT to add these secrets to `MAHE-CI-D0C-PRD-KV01` :

Secret name	Description
DATABASE-URL	Full PostgreSQL connection string to <code>eduverify</code>
JWT-SECRET	Random 64+ character hex (generate new for Prod — do not reuse UAT)
SEED-ADMIN-PASSWORD	Strong production admin password for ADM-001
AZURE-STORAGE-CONNECTION-STRING	Prod storage account connection string (if not using MI only)

Generate JWT secret (run once on secure machine):

```
node -e "console.log(require('crypto').randomBytes(48).toString('hex'))"
```

Confirm managed identity `97fb7c66-1571-43bb-96dd-814578216a4f` has **Key Vault Secrets User** on the vault.

7. Phase D — Configure production App Service

Portal → `MAHE-CI-D0C-PRD-APP01` → **Environment variables** → Add all settings → **Apply** → **Restart**.

Name	Production value
NODE_ENV	production
PORT	8080
AUTO_SETUP	true
STORAGE_PROVIDER	azure
AZURE_STORAGE_ACCOUNT_NAME	mahecidocprdsgacct01
AZURE_STORAGE_CONTAINER	eduverify-documents
AZURE_STORAGE_CONNECTION_STRING	From Key Vault (prod storage only)
DATABASE_URL	From Key Vault (prod PostgreSQL / eduverify)
JWT_SECRET	From Key Vault (new for prod)
JWT_EXPIRES_IN	12h
SEED_ADMIN_ID	ADM-001

SEED_ADMIN_NAME	Verification Cell Admin
SEED_ADMIN_PASSWORD	From Key Vault (strong — not ChangeMe@2026)
CORS_ORIGIN	https://maheblreduverify.manipal.edu
PG_POOL_MAX	25
WEBSITE_NODE_DEFAULT_VERSION	~20

Deployment slot setting: Leave **unchecked** unless using staging slots.

CORS_ORIGIN must match the exact public URL students use (custom domain after gateway).

- 1 Copy **Default domain** from Prod App Service Overview (may be regional, e.g.centralindia-01.azurewebsites.net). Use this for initial smoke tests before custom domain is live.

8. Phase E — Deploy application code to Production

Option 1 — GitHub Actions (recommended)

- 1 GitHub repo `mahe-eduverify` → **Settings** → **Secrets** → **Actions**.
- 2 Add secret `AZURE_WEBAPP_PUBLISH_PROFILE` — download from **MAHE-CI-DOC-PRD-APP01** → Overview → **Download publish profile** (enable SCM basic auth if blocked).
- 3 Update `.github/workflows/mahe-prod-deploy.yml` → set `AZURE_WEBAPP_HOST` to Prod App Service **Default domain** from Overview.
- 4 GitHub → **Actions** → **Deploy MAHE Production** → **Run workflow** → type `deploy-prod` to confirm.

Option 2 — Azure CLI from developer machine

```
az login
az account set --subscription "MAHEBLR-Admission"

cd eduverify-server
az webapp deploy \
  --resource-group MAHE-CI-DOC-PRD-RG01 \
  --name MAHE-CI-DOC-PRD-APP01 \
  --src-path . \
  --type zip
```

9. Phase F — Smoke test (before custom domain)

- 1 Open `https://<prod-default-domain>/api/health` → expect `{"ok":true,"db":"up"}`.
- 2 Verification Cell Login → `ADM-001` + production `SEED_ADMIN_PASSWORD`.
- 3 Upload **one** test student; upload one PDF; confirm blob in `mahecidocprdstgacct01 / eduverify-documents`.

- 4 Delete test student after smoke test (supervisor) if using real-looking data.

10. Phase G — Application Gateway + custom domain

Coordinated with MAHE infrastructure team after app smoke test passes.

Step	Owner	Action
1	IT	Provision Application Gateway in front of MAHE-CI-DOC-PRD-APP01
2	IT	Attach existing wildcard SSL certificate for manipal.edu
3	IT	DNS: <code>maheblreduverify.manipal.edu</code> → Application Gateway
4	App team	Update <code>CORS_ORIGIN</code> to <code>https://maheblreduverify.manipal.edu</code>
5	App team	Restart App Service; test health and login on custom domain
6	IT	Restrict public access (private endpoints) per target architecture

11. Phase H — Production data load (5,000–10,000 students)

- 1 Sign in as supervisor ADM-001 on production URL.
- 2 **Bulk Upload** → download template → prepare student Excel from admissions data.
- 3 Upload in batches of **1,000–2,000** rows; validate each batch before next.
- 4 Generate verification schedule → allocate students (orientation week).
- 5 Communicate student URL and login instructions (app number + DOB for first login).

12. Security checklist (production)

#	Item	Done
1	JWT_SECRET is unique, 32+ chars, not UAT value	☐
2	SEED_ADMIN_PASSWORD is strong; not default	☐
3	CORS_ORIGIN is exact production URL (not *)	☐
4	Blob container is private (no public read)	☐
5	Secrets in Key Vault; not in git	☐
6	Storage keys rotated after any exposure in email/chat	☐
7	HTTPS only on public URL	☐
8	Application Insights enabled	☐
9	UAT and Prod use separate DB and storage	☐
10	Admin accounts: disable unused verifiers	☐

13. Rollback plan

Scenario	Action
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Bad code deploy	GitHub → re-run previous successful workflow commit; or App Service → Deployment Center → redeploy previous package
Database issue	Restore PostgreSQL from Azure backup (point-in-time); contact IT
Gateway/DNS issue	Revert DNS to maintenance page; IT rolls back gateway config
Critical bug at go-live	Disable student access via gateway; keep admin access for fixes; communicate delay to admissions

14. Post go-live monitoring (first 72 hours)

- Monitor `/api/health` every hour (automated or manual)
- Application Insights → failures, response times, exceptions
- App Service → CPU/memory; scale out if CPU > 70% sustained
- PostgreSQL → connections, storage growth
- Storage account → capacity (plan ~100 GB per 5k students)
- Verification Cell hotline for login/upload issues

15. Contacts & escalation

Role	Contact
Application SPOC	abhijit.das@manipal.edu
Application support	subin.k@manipal.edu
Azure / infrastructure	MAHE IT (subscription MAHEBLR-Admission)
Verification Cell	Admission verification team lead

16. Production timeline (suggested)

Day	Activity
D-7	UAT sign-off completed
D-5	Prod PostgreSQL + storage + Key Vault secrets ready
D-4	Prod App Service env vars + code deploy + smoke test
D-3	App Gateway + SSL + DNS to maheblreduverify.manipal.edu
D-2	Bulk upload pilot batch (100 students) on Prod
D-1	Full bulk upload; verifier training; schedule allocation

D-Day	Students go live; war-room monitoring
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End state: Students access <https://maheblreduverify.manipal.edu> → login with application number + DOB → upload documents. Staff use Verification Cell portal. Metadata in PostgreSQL `eduverify`; files in Blob `eduverify-documents`. UAT environment remains for future releases.